TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Software for Wet Processing

Recipe and dosing software for the dyehouse: how a shade recipe travels from the colour kitchen to the machine controller — converted, weighed, dosed and recorded by software — so that 500 kg of fabric gets exactly the chemistry the lab approved for 10 grams.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 6 · LEVEL 3 TERM 2

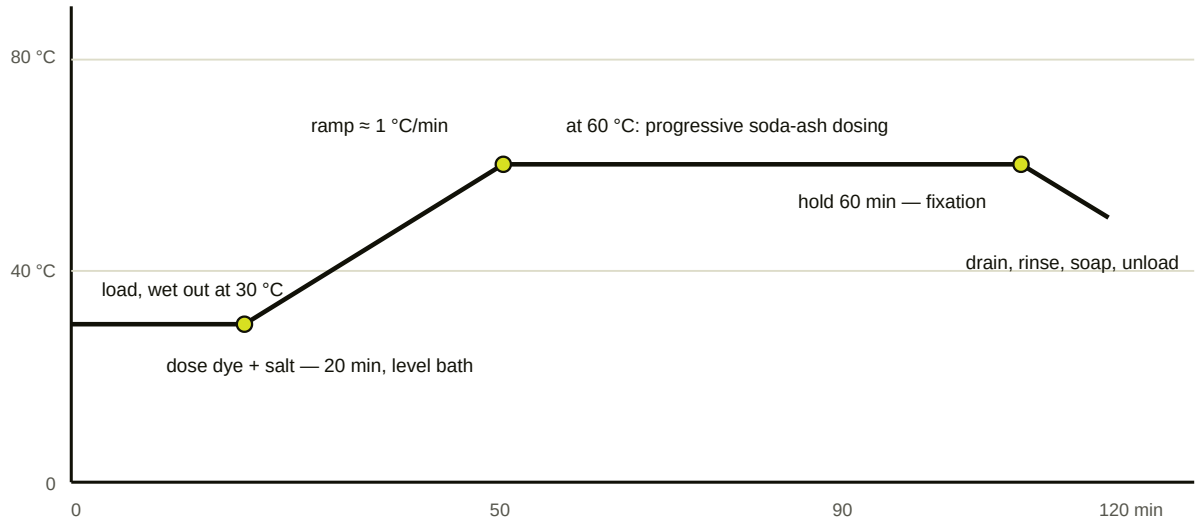
1. Aim

To build a bulk dyeing recipe in wet-processing software from a lab recipe: converting between % owf and g/L at a stated liquor ratio, computing every chemical quantity for the batch, generating the dosing sequence, and costing the batch — checking the software's figures by hand.

2. Concepts you need

- **Two languages of recipe:** dyes are stated in **% owf** (on weight of fabric — a percentage of the goods), most auxiliaries in **g/L** (grams per litre of bath). The bridge between them is the **liquor ratio (LR)**: bath volume = fabric weight × LR. Nearly every recipe error in a dyehouse is a conversion error between these two languages; the software's first job is to make conversion automatic and checked.
- **Recipe management:** the approved lab recipe (from the lab-dip stage) is stored once in a recipe database — dyes, auxiliaries, the temperature–time program, the fibre and the shade reference. Bulk production calls the recipe up and scales it to the machine load; nobody re-types it, so nobody can transpose a digit.
- **Automatic dispensing and dosing:** liquid chemicals are held in day tanks and dosed by metered pumps under program control — the right volume, at the programmed point of the curve (Figure 1), including progressive alkali dosing that a man with a bucket cannot reproduce. Powder dyes are weighed in the colour kitchen, increasingly with software-guided weighing that refuses the wrong product or an out-of-tolerance weight.
- **The machine controller** executes the temperature–time program: ramps, holds, fills, drains, dosing calls. Connected to the central system (Experiment 2's SCADA level), it records the actual curve of every batch — the evidence when a shade dispute arises.

- **Stock and cost:** the same system decrements chemical stock by what each batch consumed and costs the batch — dyes are often the largest single cost in a dyeing, and a recipe that cannot be costed cannot be quoted.
- **Right-first-time:** the dyehouse performance figure — the percentage of batches passing shade without addition or stripping. Recipe discipline through software is the main lever on it; every re-dye doubles water, energy and time for that batch.



e programmed temperature–time curve of the worked example (reactive dyeing of cotton). The controller, not the operator's memory, calls each dosing event at its poin

3. Lab procedure

1. Call up the lab recipe in the software: cotton knit, reactive dyes, shade "Deep Red", lab recipe — dye 1.5 % owf, Glauber's salt 60 g/L, soda ash 20 g/L, wetting agent 1 g/L, sequestering agent 2 g/L.
2. Create the bulk job: fabric load 500 kg, liquor ratio 1:8. Let the software scale the recipe and produce the batch sheet.
3. Independently hand-calculate every quantity (worked example) and reconcile against the batch sheet line by line. Any disagreement is resolved before proceeding — state which figure was wrong and why.
4. Program the dosing sequence: dye stock as a 10 % solution dosed over 10 minutes; salt in two portions; soda ash progressive over 30 minutes at 60 °C (Figure 1). Record the pump rate the software sets for the dye stock.
5. Cost the batch from the system's price file and record the chemical cost per kg of fabric.
6. Save the job, then re-open it as a repeat order ("re-dye same shade, 300 kg") and verify that the system rescales quantities while keeping % owf and g/L unchanged.

4. Worked example — scaling the lab recipe to a 500 kg batch

Bath volume = 500 kg × LR 8 = **4,000 L**

ITEM	RECIPE STATEMENT	BATCH QUANTITY — CALCULATION
Reactive dye, Deep Red	1.5 % owf	500 × 1.5 ÷ 100 = **7.5 kg**
Glauber's salt	60 g/L	60 × 4,000 = 240,000 g = **240 kg**
Soda ash	20 g/L	20 × 4,000 = **80 kg**
Wetting agent	1 g/L	1 × 4,000 = **4 kg**
Sequestering agent	2 g/L	2 × 4,000 = **8 kg**

Dosing: the dye is dispensed as a 10 % stock solution, so the dosing system must deliver 7.5 ÷ 0.10 = **75 kg ≈ 75 L** of solution. Spread over the programmed 10-minute dosing window, the pump rate is 75 ÷ 10 = **7.5 L/min** — a figure the operator can check against the dosing tank's sight glass as the batch runs.

Cost: at Tk 1,200 per kg, the dye alone costs 7.5 × 1,200 = **Tk 9,000**, i.e. 9,000 ÷ 500 = **Tk 18 per kg of fabric** — before salt, alkali, water, steam and time. This is the number merchandising needs, and it falls out of the batch sheet automatically *only if* the quantities above are right: a single misplaced decimal in the salt line (24 kg instead of 240 kg) would not change the dye cost, but it would change the shade — exhaust dyeing with too little electrolyte gives a pale, unlevel batch and a full re-dye.

5. Practice tasks

1. Scale the same recipe system for a smaller machine: 200 kg of fabric at LR 1:10, with dye at 2.0 % owf, salt 40 g/L and soda ash 15 g/L. Give the bath volume and all three quantities.
2. For task 1, the dye costs Tk 950/kg. Calculate the dye cost of the batch and the dye cost per kg of fabric.
3. An operator proposes entering the soda ash "all at once at the start, to save time" instead of progressive dosing at 60 °C. State, in process terms, why the software's programmed dosing protects the shade.

Answer key

1. Bath volume = 200 × 10 = **2,000 L**. Dye = 200 × 2.0 ÷ 100 = **4.0 kg**. Salt = 40 × 2,000 = **80 kg**. Soda ash = 15 × 2,000 = **30 kg**.

2. Dye cost = 4.0 × 950 = **Tk 3,800**; per kg of fabric = 3,800 ÷ 200 = **Tk 19/kg**. (Higher than the worked example's Tk 18/kg because the shade is deeper — 2.0 % against 1.5 % owf — even though the dye itself is cheaper per kg.)
3. Reactive dye fixes to cotton when alkali raises the bath pH. Dumping all the soda ash at the start raises pH immediately, so dye strikes and fixes rapidly at the first contact points — before it has levelled through the load — giving unlevel, patchy dyeing, and more dye is lost to hydrolysis in the bath instead of fixing evenly. Progressive dosing at the dyeing temperature lets the dye first exhaust and migrate evenly under salt, then fixes it gradually and uniformly. The programmed curve is a quality device, not a convenience.
